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A AIML PROJECT REPORT

on

"AI Sematic Skill Matching and Gap Analyzer"

Submitted by

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2025 - 26

ABSTRACT

AI Semantic Skill Matching and Gap Analyzer is an AI-based system developed to compare candidate skills with job requirements using Natural Language Processing (NLP) and Machine Learning techniques. The system extracts skills from resumes and job descriptions, performs semantic analysis to measure skill similarity, identifies missing skills, and provides recommendations for improvement. It helps recruiters improve hiring accuracy and supports students and professionals in enhancing their career opportunities through intelligent skill gap analysis.

AI Semantic Skill Matching and Gap Analyzer is an intelligent recruitment support system that uses Artificial Intelligence (AI) and Natural Language Processing (NLP) to evaluate candidate skills against job requirements. The system analyzes resumes and job descriptions, measures skill similarity, detects missing competencies, and suggests areas for improvement. By automating the skill evaluation process, the project reduces manual effort, improves recruitment efficiency, and helps users strengthen their professional skills for better career opportunities.

Keywords:

Artificial Intelligence (AI), Natural Language Processing (NLP), Skill Matching, Skill Gap Analysis, Resume Analysis, Machine Learning, Semantic Analysis, Recommendation System, Recruitment Automation, Career Development.

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CHAPTER 1

INTRODUCTION

1.1 Project Overview

The introduction of the AI Semantic Skill Matching and Gap Analyzer provides a clear understanding of how Artificial Intelligence (AI) and Natural Language Processing (NLP) can improve modern recruitment and career guidance systems. It explains the drawbacks of traditional keyword-based recruitment methods and emphasizes the need for semantic analysis to identify skills more accurately. The introduction also outlines the main objectives of the project, such as skill extraction, semantic matching, gap identification, and recommendation generation. Additionally, it highlights the benefits of the system for both recruiters and job seekers by improving hiring accuracy, reducing manual effort, and supporting career development. Overall, the introduction effectively presents the purpose, scope, and significance of the project.

The system extracts skills from candidate resumes, compares them with industry or job requirements, calculates a matching score, and provides recommendations for improvement. It helps recruiters reduce manual screening effort and assists students and professionals in identifying the skills needed for better career opportunities. The project combines Machine Learning, NLP, and data analysis techniques to create an efficient and intelligent skill evaluation platform.

1.2 Problem Statement

Traditional recruitment systems mainly rely on keyword-based resume screening methods, which often fail to accurately identify candidate skills and competencies. These systems cannot effectively understand semantically related skills, resulting in unsuitable candidate selection and missed opportunities for qualified applicants. Manual resume evaluation is also time-consuming, less efficient, and difficult to manage when handling a large number of applications.

Additionally, n addition, many students and professionals are unaware of the specific skills required for different job roles and industries. They often struggle to identify their strengths, weaknesses, and missing competencies needed for career growth.

Existing systems provide limited guidance for skill improvement and career development. Traditional recruitment systems rely mainly on keyword-based resume screening, which may not accurately identify candidate skills and job suitability. Manual resume evaluation is time-consuming and less efficient when handling large numbers of applications. Candidates also face difficulty in identifying missing skills required for specific job roles. Therefore, an AI-based system is needed to perform semantic skill matching, identify skill gaps, and provide recommendations for career improvement.

1.3 Proposed Solution

The proposed solution is an AI-based Semantic Skill Matching and Gap Analyzer that uses Natural Language Processing (NLP) and Machine Learning techniques to analyze resumes and job descriptions intelligently. The system extracts candidate skills, qualifications, and job requirements, then performs semantic matching to identify the similarity between them.

The system calculates a skill matching score, detects missing or weak skills, and provides personalized recommendations for improvement. It also generates reports and visual dashboards to help recruiters and candidates understand the analysis easily. This solution reduces manual recruitment effort, improves hiring accuracy, and supports users in enhancing their professional skills and career opportunities.

1.4 Project Objectives

- To develop an AI-based skill matching system.
- To analyze resumes and job descriptions using NLP techniques.
- To extract candidate skills and qualifications automatically.
- To perform semantic skill matching between candidates and job roles.
- To identify skill gaps and missing competencies.
- To provide recommendations for skill improvement.
- To improve recruitment accuracy and efficiency.
- To reduce manual effort in resume screening.
- To assist students and professionals in career development.
- To generate reports and visual dashboards for analysis.

CHAPTER 2

SYSTEM REQUIREMENT SPECIFICATIONS

2.1 Technical Architecture

The technical architecture of the AI Semantic Skill Matching and Gap Analyzer consists of a user interface layer, application layer, data processing layer, semantic matching engine, gap analysis module, recommendation system, database layer, and visualization layer. The user interface allows users to upload resumes and job descriptions and view analysis results. The application layer manages communication between frontend and backend components. The data processing layer uses Natural Language Processing (NLP) to extract skills, qualifications, and keywords from documents. The semantic matching engine compares candidate skills with job requirements using Machine Learning algorithms to calculate matching scores. The gap analysis module identifies missing skills, while the recommendation system suggests improvements and learning paths. The database layer stores resumes, job descriptions, and analysis reports using databases like MongoDB or MySQL. Finally, the visualization layer presents dashboards, charts, and reports for easy understanding of the results.

Information Extraction on Technical Architect:

The technical architecture of the AI Semantic Skill Matching and Gap Analyzer includes a user interface, application layer, data processing layer, semantic matching engine, gap analysis module, recommendation system, database layer, and visualization layer. The system extracts skills and qualifications from resumes and job descriptions using NLP techniques, compares them semantically, identifies skill gaps, and provides recommendations. The database stores analysis data, while dashboards and reports help users easily understand the results

2.2 Technology Stack

- **Frontend Technologies:**

  HTML: Used to create the structure of web pages.CSS: Used for designing and styling the user interface.JavaScript: Used to add interactivity and dynamic featuresChart.js: Used to display dashboards, graphs, and visual reports..

- **Backend Technologies:**
  Python: Main programming language used for application development.

  Flask / Django: Frameworks used to manage server-side operations and application logic.

  Machine Learning: Used for skill matching and prediction analysis.

  Natural Language Processing (NLP): Used for resume parsing, skill extraction, and semantic analysis.

  MongoDB / MySQL: Databases used to store resumes, job descriptions, and analysis data.

  Libraries: Pandas, NumPy, NLTK, and Scikit-learn are used for data processing and AI functionalities

- **Visualization Tools:**

  The AI Semantic Skill Matching and Gap Analyzer uses visualization tools such as Chart.js, Power BI, and Matplotlib to generate dashboards, charts, graphs, and analytical reports. These tools help users and recruiters easily understand skill matching scores, gap analysis results, and recommendations through visual representation.

- **Development Tools:**
  Development tools such as VS Code, Jupyter Notebook, GitHub, and Postman are used for coding, testing, debugging, AI model development, version control, and API integration. These tools support efficient project development and system management.

- **Compatibility:**
  The system is compatible with operating systems like Windows, Linux, and macOS and supports modern web browsers such as Chrome, Firefox, and Edge. It can be accessed through desktops, laptops, and mobile devices, making the platform flexible and user-friendly.

2.3 Dataset Used

The AI Semantic Skill Matching and Gap Analyzer uses datasets containing resumes, job descriptions, candidate skills, qualifications, and industry job requirements. The dataset includes information such as technical skills, educational qualifications, work experience, certifications, and job role requirements. Public datasets from platforms like Kaggle or manually collected sample resumes and job postings can be used for training and testing the system. The dataset helps in performing skill extraction, semantic matching, and gap analysis using AI and NLP techniques.

2.4 Techniques Used

The AI Semantic Skill Matching and Gap Analyzer uses various AI and NLP techniques to analyze resumes and job descriptions effectively. Natural Language Processing (NLP) techniques are used for text preprocessing, tokenization, skill extraction, and semantic analysis. Machine Learning algorithms are applied for skill matching and prediction analysis. Semantic similarity techniques help compare candidate skills with job requirements accurately. Resume parsing and keyword extraction methods are used to identify important information from documents, while data visualization techniques are used to present analysis results through dashboards and reports.

CHAPTER 3

RESULTS AND DISCUSSIONS

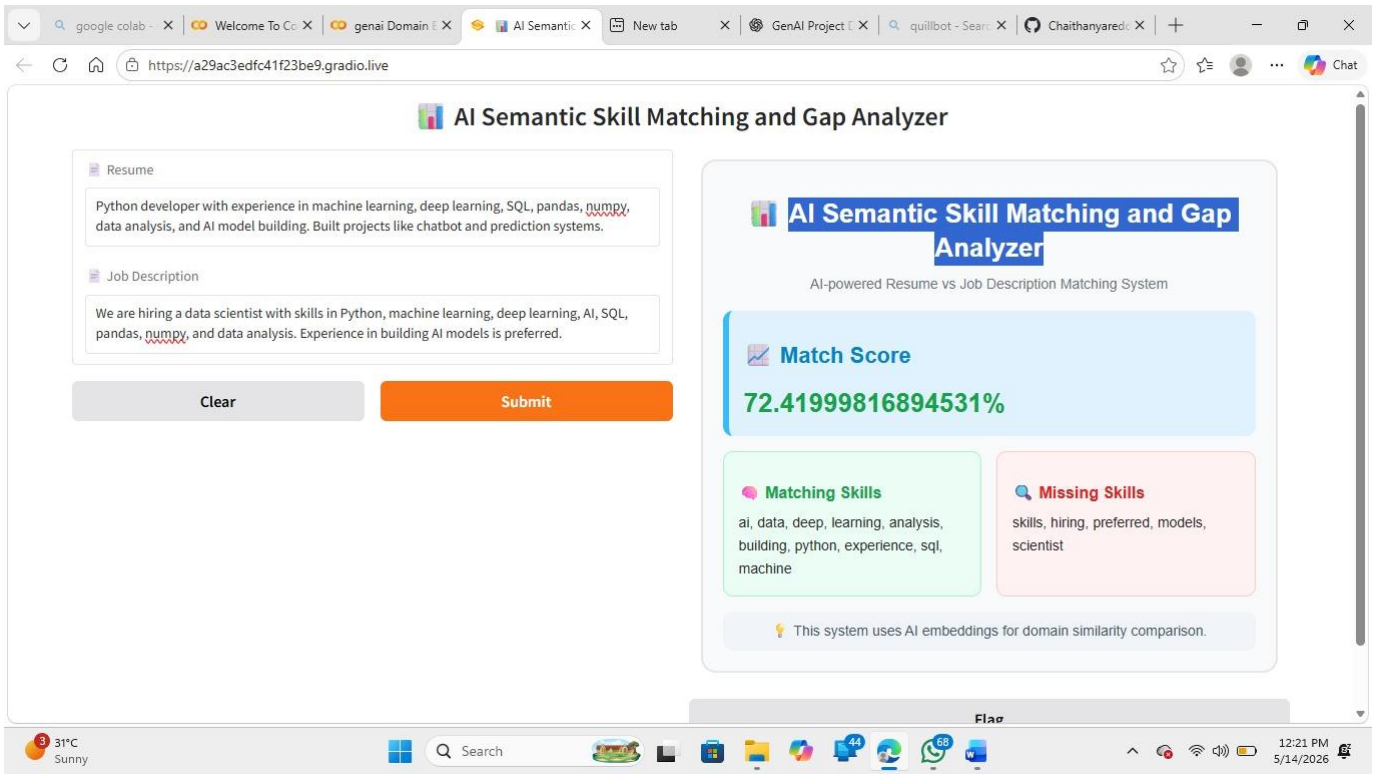

Fig. 3.1: AI Sematic Skill Matching and Gap Analyzer

Figure 3.1 The image shows the user interface of the AI Semantic Skill Matching and Gap Analyzer system developed for resume and job description analysis. The application allows users to enter resume details and job requirements, then compares them using AI-based semantic analysis techniques. The system generates a match score of approximately 72%, displays matching skills such as Python, SQL, and machine learning, and identifies missing skills required for the job role. The interface also includes features like skill visualization, gap analysis, and AI embedding-based similarity comparison to improve recruitment and career guidance processes.

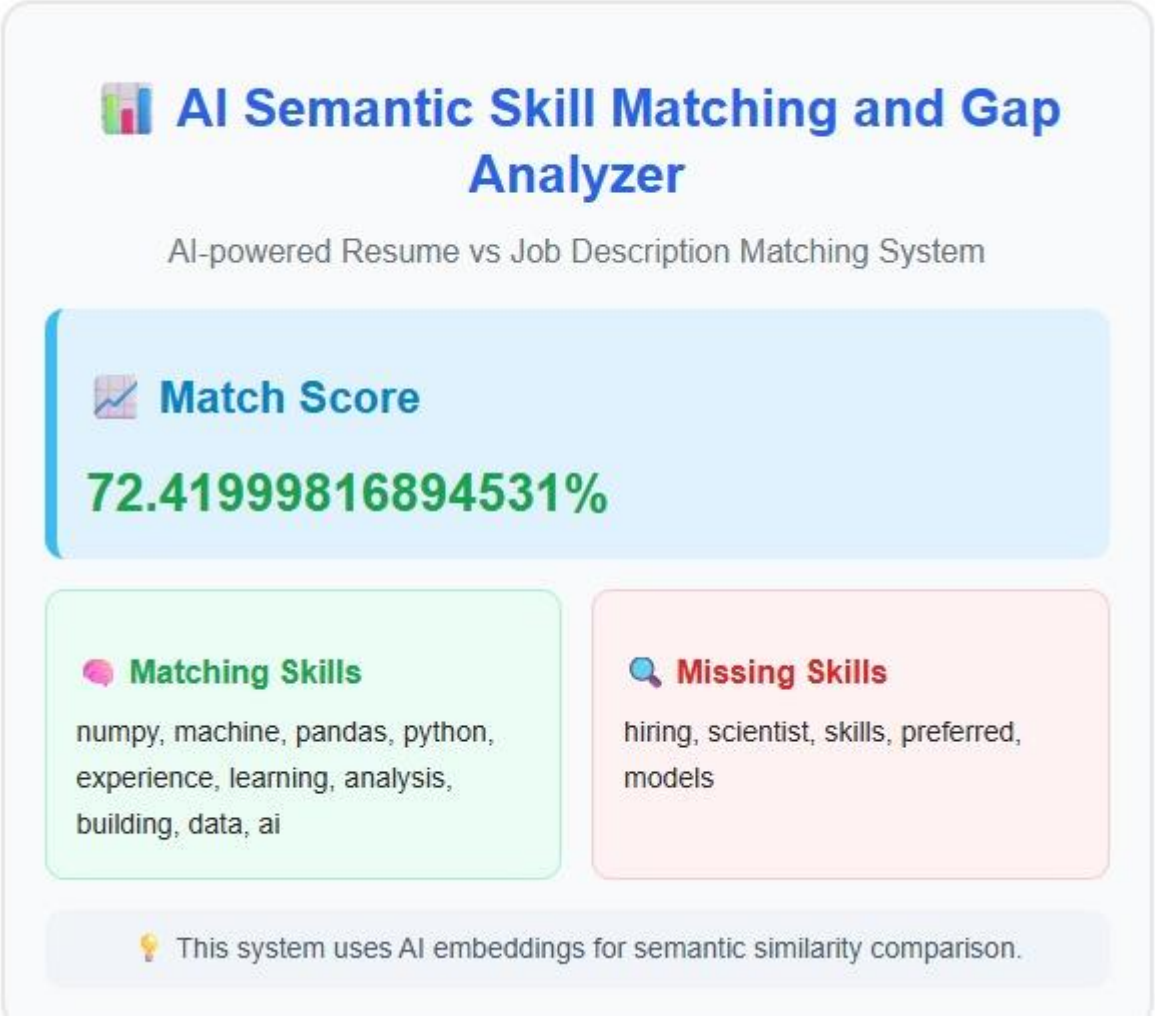

Fig. 3.2: AI Sematic Skill Matching and Gap Analyzer Dashboard

The Figure 3.2 This figure shows an AI Semantic Skill Matching and Gap Analyzer system used for comparing resumes with job descriptions.The dashboard calculates a match score of 72.42%, indicating how well the candidate's skills fit the job requirements.It displays matching skills such as Python, NumPy, Pandas, Machine Learning, Data Analysis, and AI.The system also identifies missing skills like hiring knowledge, scientist-related skills, preferred qualifications, and models.The application uses AI embeddings and semantic similarity techniques to analyze and compare resume content intelligently.

CHAPTER 4

CONCLUSION AND FUTURE SCOPE

The AI Semantic Skill Matching and Gap Analyzer is an intelligent system developed to improve the recruitment and career development process using Artificial Intelligence (AI) and Natural Language Processing (NLP). The project successfully analyzes resumes and job descriptions, extracts important skills and qualifications, performs semantic skill matching, and identifies missing competencies. Unlike traditional keyword-based recruitment systems, this project understands the meaning and relationship between skills, resulting in more accurate candidate evaluation and selection.

The system helps recruiters reduce manual effort and save time during resume screening while improving hiring accuracy and efficiency. It also assists students and professionals in understanding their strengths and weaknesses by providing personalized recommendations for skill improvement. The integration of Machine Learning, NLP, and data visualization techniques makes the system efficient, scalable, and user-friendly. Overall, the project demonstrates how AI-driven solutions can enhance modern recruitment systems and support career growth through intelligent skill analysis.

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